

what that computer can achieve in terms of display. The same applies to hard drives and other components. Networking has reduced the limitations of storage space—now other components of the computer need to break free.

In a Utility Computing (UC) scenario, nothing is bound to anything else. There is a virtual pool of resources, available to any task, and as a result, any user, who needs it. The dynamics of this resource pool are constantly changing as the needs of the collective user-base change.

In order to understand the functioning here, imagine a PC just the way it is, with the difference being the fact that none of the components in your computer belong solely to you. In a UC environment, your computer's unused resources could be allocated to other users in the same UC grid. It all boils down to leasing computing resources just as we do for electricity. Billing can also be done easily on a per-grid, per-group or per-person basis. We could have next-generation *Params* sitting in government labs, computing for the masses.

What About Software?

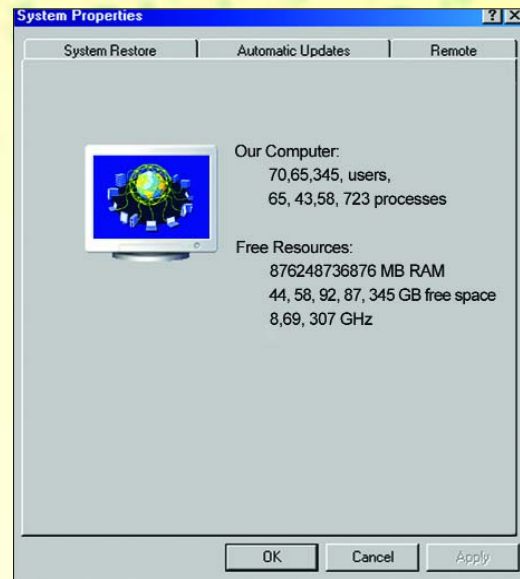
Current software is as limited as our hardware, but this will need to change in order to achieve Utility Computing. In a true UC environment, things such as an OS don't mean a thing. Users are not classified according to 'Windows' or 'Unix', but merely by the task they want done. Yes, applications need to remain familiar, but hardware and software need to become two independent entities.

So You Mean Grid Computing?

No, we do not! Grids may simply enable utility computing. Grid computing is used to achieve impressive number crunching capabilities, as is the case with the SETI@home project.

The Search for Extra-Terrestrial Intelligence (SETI) revolves around capturing radio waves from space and computing them to look for patterns—as in communication. The problem is that all solar masses in the universe release radio waves into space. Essentially, the SETI project is looking for the proverbial needle in a haystack the size of our universe. Since they don't have multi-billion dollar research grants, they hit upon the idea of using the Internet as a distribution tool and getting computing help from the masses. If you

As technology becomes more pervasive than ever, your PC might soon wield more power than today's supercomputers



Soon, My Computer will become 'Our Computer'

download and install the SETI@home client, your computer will download a small chunk of radio data from SETI, and when you are not using your computer, it processes the data.

Because of the millions of people running this program, SETI benefits from free computing that would otherwise cost them millions of dollars in supercomputer time.

'Our' Computer?

Utility Computing, however, is not just about number crunching. It is about sharing every resource that a computer currently uses. And more importantly, UC is all about 'on-demand' resources.

In the ideal UC scenario, terms such as speed and capacity do not exist. When a task needs to be completed, it is done as fast as the overall UC grid will allow. So, say you want to render a two-hour animation you created, you don't need to worry about which drive to store the large file on, or shutting down applications to make the process faster. Hard disk space and CPU and GPU resources would be allocated as the need arises. Gigahertz clock speeds and GBs of disk space would mean nothing. A system specification



A Chip That Computes 10 Times Faster Than Existing CPUs

IBM, Toshiba and Sony recently collaborated on research for a new chip called the 'Hard Cell' or 'Cell'. This new chip will power Sony's upcoming PS3 gaming console. Toshiba will also put the Cell to use in an upcoming line of HDTVs. The companies call it a "supercomputer on a chip."

The best thing about the Cell is that it is rated to be 10 times faster than existing CPUs. According to IBM, it contains "eight synergistic processors and top clock speeds of greater than 4 GHz (as measured during initial hardware testing)." This new chip is capable of massive floating point processing—which is required by graphics.

However, that's not the reason why we mention it here. The feature that excites us most is that the cell is a multi-core chip with a difference: each processing unit in the Cell can process information from different tasks. Existing chips can only handle one

task at a time, so even if the task is relatively small—using almost nothing of the CPU's capability—the processor does not accept more tasks. With the Cell, however, multiple tasks can be executed simultaneously, with the central Processing Unit (PU) deploying work to the Attached Processing Units (APUs). This increases efficiency tremendously, and is the reason why the creators are confident of the Cell outperforming current chips 10 to 1. Since the APUs are sort of like freelance workers, unused computing power can be shared over a network of Cell-based computers, forming a computing Grid.

The 'cell' that gives the chip its name doesn't refer to the hardware, but to a virtual clump of software that roams the system looking for computing resources. For Sony's PS3, this means that a network of PS3s can share resources so that everyone has a better gaming experience. This could be the beginnings of Utility Computing. At least we know now that the hardware already exists!